

Performance Analysis of Digital Beamforming mmWave MIMO with Low-Resolution DAC/ADC

F. Pasic, M. Mussbah, S. Schwarz, F. Tufvesson, M. Rupp and C. F. Mecklenbräuer

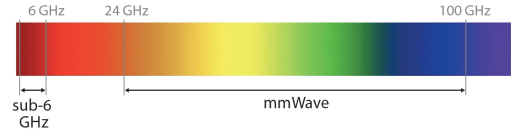
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- increasing demand for high data rate transmission

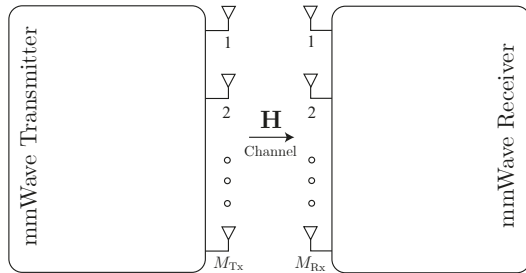
- increasing demand for high data rate transmission
- sub-6 GHz band cannot meet these requirements



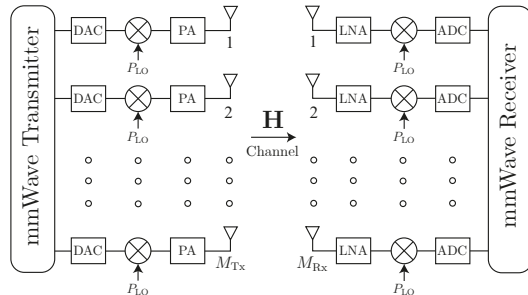
- increasing demand for high data rate transmission
- sub-6 GHz band cannot meet these requirements
- mmWave bands allow for significantly higher data rates



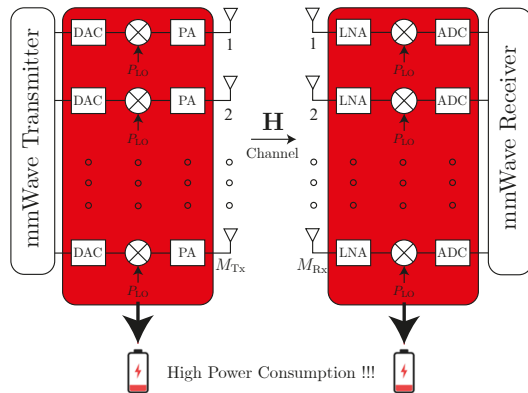
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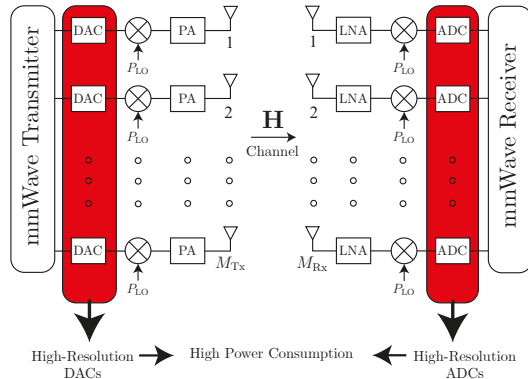
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 - low-latency applications



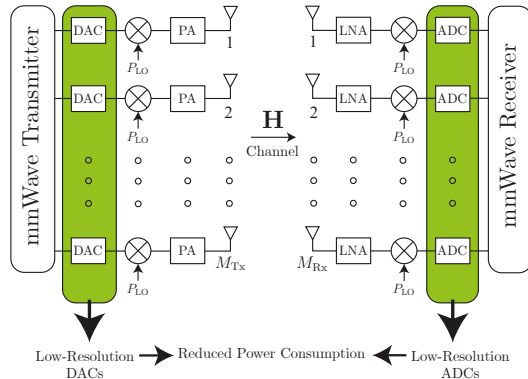
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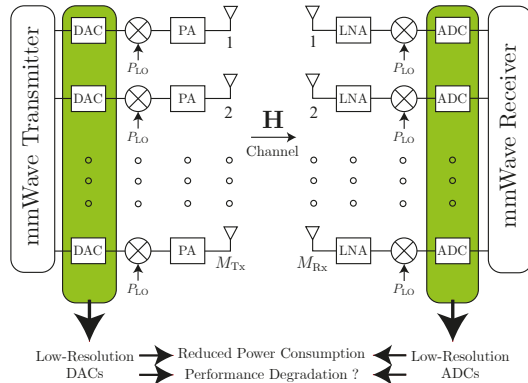
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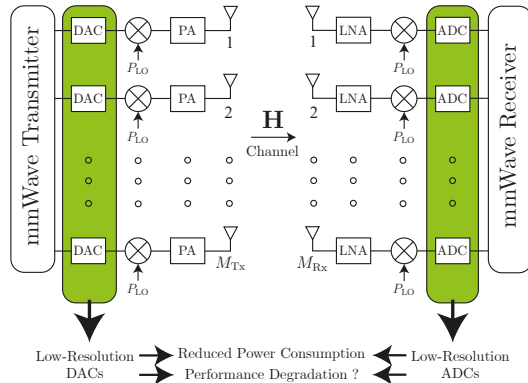
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- evaluate performance under practical hardware constraints



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 - additive quantization noise model (AQNM)
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 - additive quantization noise model (AQNM)
 - practical hardware constraints
- **evaluate system performance in terms of spectral efficiency (SE) and energy efficiency (EE)**
 - conventional and advanced channel estimation methods
 - different antenna configurations
 - varying channel conditions in terms of Rician K -factor

System Model

Simulation Results

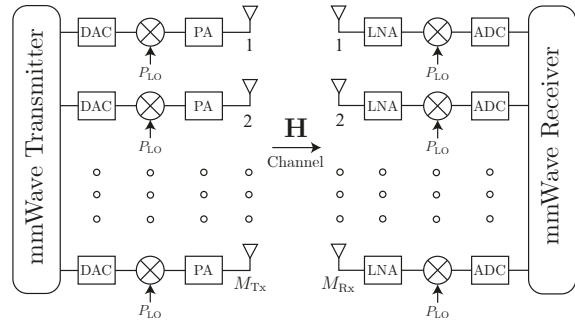
Summary and Conclusion

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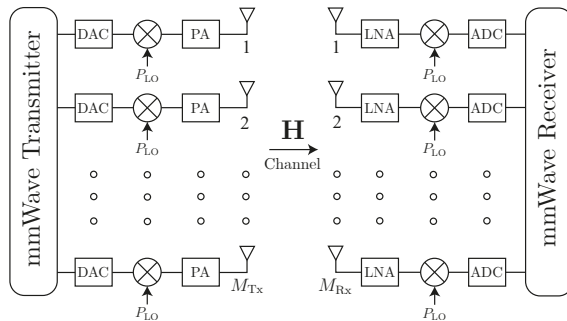
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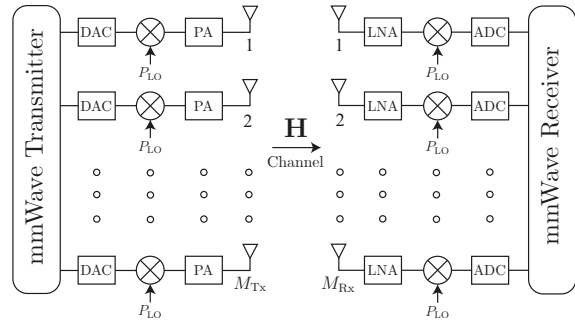
- mmWave MIMO OFDM system



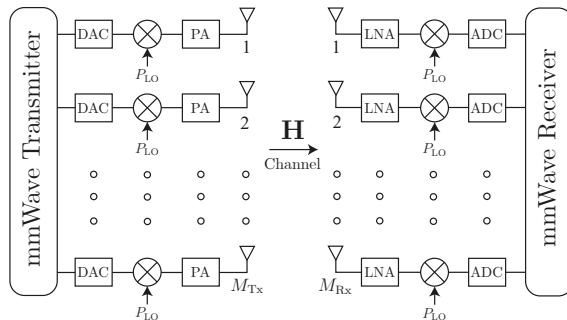
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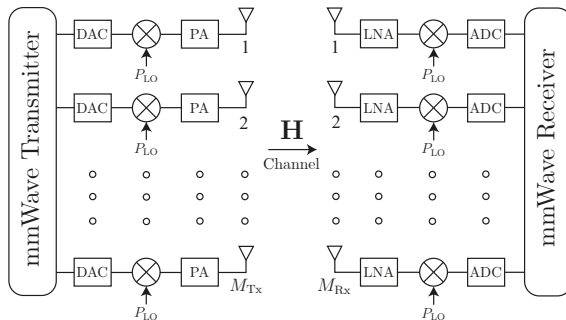


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 - reciprocal channel responses
- frequency-selective Rician channel model
 - subcarrier index n
 - variable K -factor



$$\mathbf{H}[n] = \underbrace{\sqrt{\frac{K}{1+K}} \mathbf{H}_{\text{fs}}}_{\text{free-space}} + \underbrace{\sqrt{\frac{1}{1+K}} \mathbf{H}_{\text{sp}}[n]}_{\text{stochastic part}}$$

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- link establishment
 - training phase
 - data transmission



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- data transmission
 - input-output relationship

$$\mathbf{y}[n] = (\overline{\mathbf{Q}}[n])^H \mathbf{H}[n] \left(\overline{\mathbf{F}}[n] (\overline{\mathbf{P}}[n])^{1/2} \mathbf{x}[n] + \mathbf{d}[n] \right) + (\overline{\mathbf{Q}}[n])^H \mathbf{w}[n],$$

transmit symbols $\mathbf{x}[n] \in \mathbb{C}^{\ell_{\max} \times 1}$, power loading matrix $\overline{\mathbf{P}}[n] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$,

Tx impairments $\mathbf{d}[n] \in \mathcal{CN} \left(0, \left(\sigma_{\text{RF}}^2 + \sigma_{\text{q,Tx}}^2 \right) \mathbf{I}_{M_{\text{Tx}}} \right)$,

precoding matrix $\overline{\mathbf{F}}[n] \in \mathbb{C}^{M_{\text{Tx}} \times \ell_{\max}}$, combining matrix $\overline{\mathbf{Q}}[n] \in \mathbb{C}^{M_{\text{Rx}} \times \ell_{\max}}$,

additive Gaussian noise $\mathbf{w}[n] \in \mathcal{CN} \left(0, \sigma_w^2 \mathbf{I}_{M_{\text{Rx}}} \right)$, receive symbols $\mathbf{y}[n] \in \mathbb{C}^{\ell_{\max} \times 1}$

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- quantized received symbols

$$\mathbf{y}_q[n] = (1 - \Upsilon(n_b)) \mathbf{y}[n] + \mathbf{w}_q[n]$$

inverse coding gain $\Upsilon(n_b) \approx \frac{\pi\sqrt{3}}{2} n_b^{-2}$, resolution bits n_b

additive quantization noise $\mathbf{w}_q[n] \in \mathcal{CN} \left(0, \sigma_{\text{q,Rx}}^2 \mathbf{I}_{M_{\text{Rx}}} \right)$

System Model

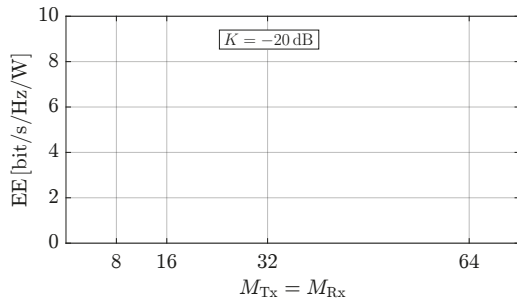
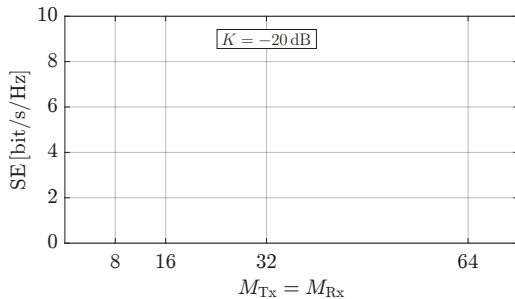
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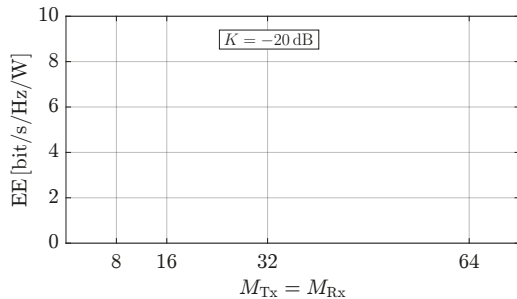
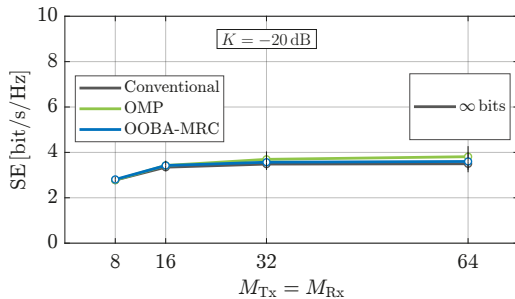
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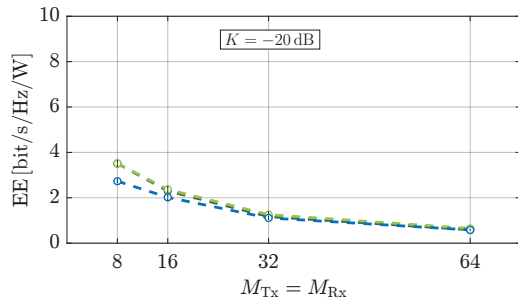
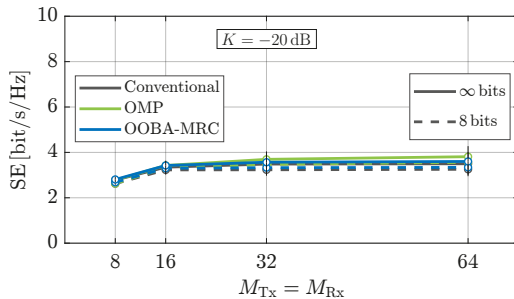
- performance evaluation
 - spectral efficiency (SE)
 - energy efficiency (EE)
 - simulation parameters
- channel estimation methods
 - conventional method
 - orthogonal matching pursuit (OMP) [1]
 - out-of-band aided maximal ratio combining (OOBA-MRC) [2]

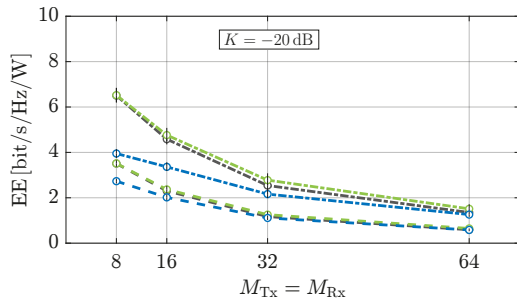
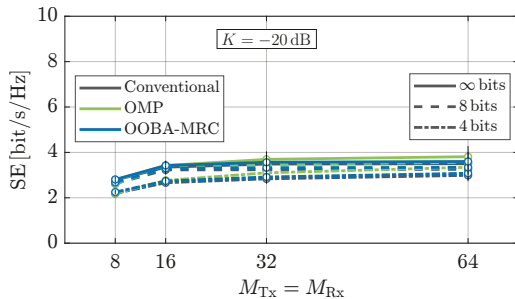
Parameter	Value
Carrier Frequency f_c	25.5 GHz
Wavelength λ	1.176 cm
Bandwidth B	403.2 MHz
Subcarrier Spacing Δf	120 kHz
Sampling Rate f_s	806.4 MHz
Cyclic Prefix t_{CP}	0.59 μ s
ADC/DAC Resolution n_b	{2, 4, 8}
Channel Model	3GPP Urban LOS

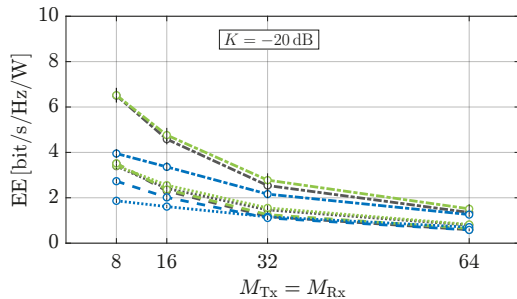
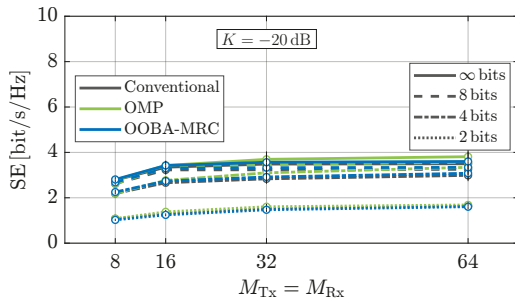
- [1] J. Lee et al., "Channel estimation via orthogonal matching pursuit for hybrid MIMO systems in millimeter wave communications," IEEE Transactions on Communications, 2016.
- [2] F. Pasic et al., "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," IEEE Transactions on Communications, 2025.

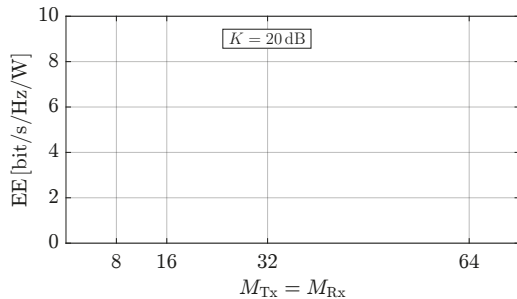
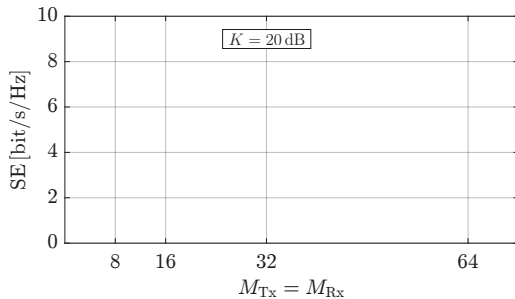


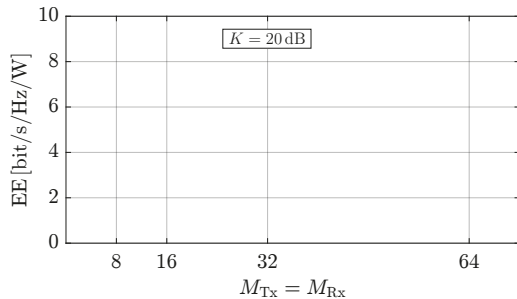
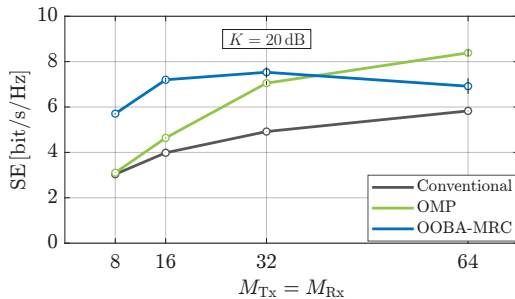


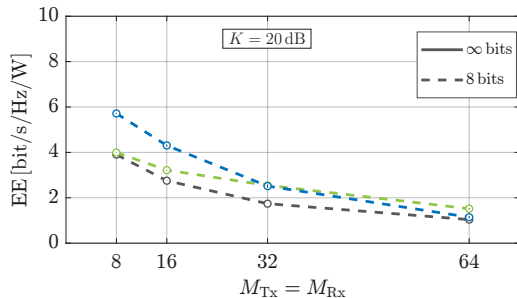
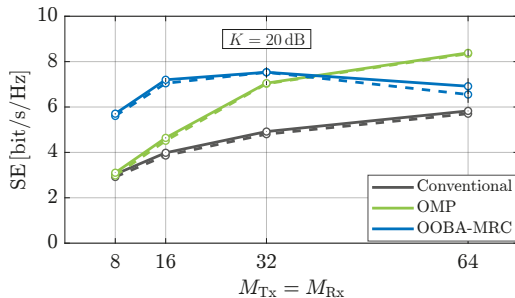


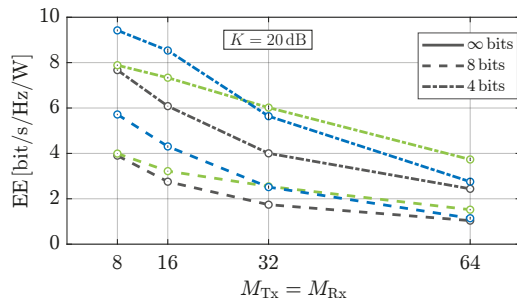
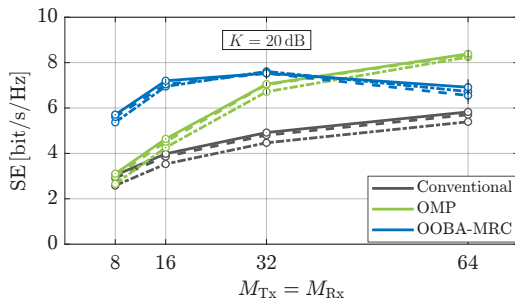


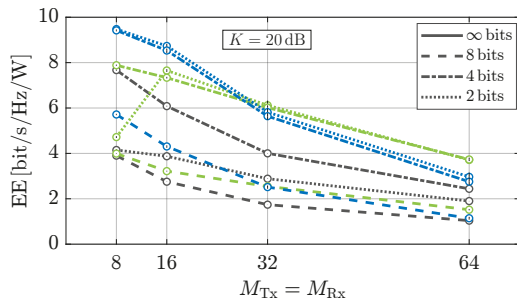
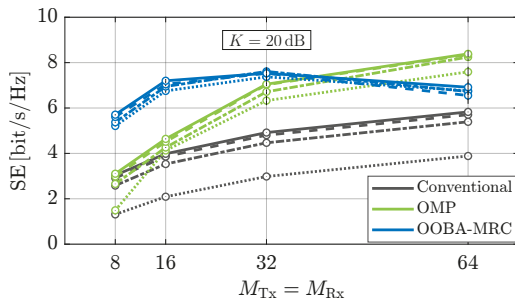












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 - advanced channel estimation methods pose increased robustness in scenarios with a high K -factor

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 - advanced channel estimation methods pose increased robustness in scenarios with a high K -factor
- **4-bit quantizers provide the highest energy efficiency**
 - favorable trade-off between power consumption and achievable data rate

Thank you for your attention!
Questions?

Faruk Pasic
faruk.pasic@tuwien.ac.at

- total power consumption

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- transmitter power consumption

$$P_{\text{Tx}} [\text{mW}] = M_{\text{Tx}} (P_{\text{DC,PA}} + P_{\text{LO}} + 2P_{\text{DAC}}),$$

power amplifier $P_{\text{DC,PA}} [\text{mW}] = \frac{1}{\eta_{\text{PA}}} \left(10^{\frac{P_{\text{out}} [\text{dBm}]}{10}} - 10^{\frac{P_{\text{in}} [\text{dBm}]}{10}} \right)$, power added efficiency η_{PA}

local oscillator (LO)/mixer P_{LO}

digital-to-analog (DAC) $P_{\text{DAC}} = 2^{n_b} f_s \text{FoM}_{\text{DAC}}$, n_b bits, sampling rate f_s , figure of merit FoM_{DAC}

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- receiver power consumption

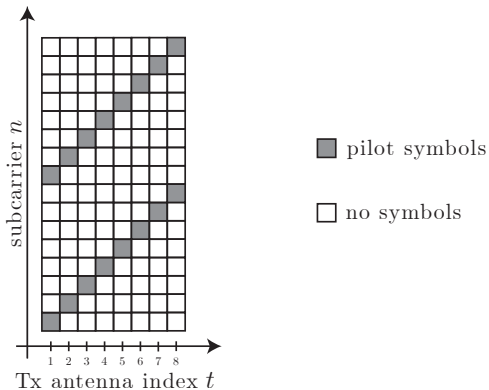
$$P_{\text{Rx}} [\text{mW}] = M_{\text{Rx}} (P_{\text{DC,LNA}} + P_{\text{LO}} + 2P_{\text{ADC}})$$

low noise amplifier (LNA) $P_{\text{DC,LNA}}$

analog-to-digital (ADC) $P_{\text{ADC}} = 2^{n_b} f_s \text{FoM}_{\text{ADC}}$, n_b bits, sampling rate f_s , figure of merit FoM_{ADC}

- training phase

- initial channel estimate: $\tilde{\mathbf{H}}[n] \implies$ resulting channel estimate: $\overline{\mathbf{H}}[n]$



- alternative channel matrix representation

$$\mathbf{H}[n] = \underbrace{\mathbf{A}_{\text{Rx}} \mathbf{H}_a[n] (\mathbf{A}_{\text{Tx}})^H}_{\text{quantized channel}} + \underbrace{\mathbf{E}[n]}_{\text{error}}$$

array responses $\mathbf{A}_{\text{Tx}} = [\mathbf{a}_{\text{Tx}}(\vartheta_1) \cdots \mathbf{a}_{\text{Tx}}(\vartheta_G)] \in \mathbb{C}^{M_{\text{Tx}} \times G}$, $\mathbf{A}_{\text{Rx}} = [\mathbf{a}_{\text{Rx}}(\varphi_1) \cdots \mathbf{a}_{\text{Rx}}(\varphi_G)] \in \mathbb{C}^{M_{\text{Rx}} \times G}$
sparse channel matrix $\mathbf{H}_a[n] \in \mathbb{C}^{G \times G}$ with its vectorized form $\mathbf{h}_a[n] = \text{vec}(\mathbf{H}_a[n])$

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- compressed sensing (CS) channel estimation

$$\bar{\mathbf{h}}_a[n] = \arg \min_{\mathbf{h}_a} \|\mathbf{y}[n] - \mathbf{\Psi} \mathbf{h}_a[n]\|_2$$

observation vector $\mathbf{y}[n] = \text{vec}((\mathbf{U}_{\text{Rx}})^H \tilde{\mathbf{H}}[n] \mathbf{U}_{\text{Tx}})$, DFT matrices $\mathbf{U}_{\text{Tx}} \in \mathbb{C}^{M_{\text{Tx}} \times M_{\text{Tx}}}$,

$\mathbf{U}_{\text{Rx}} \in \mathbb{C}^{M_{\text{Rx}} \times M_{\text{Rx}}}$

sensing matrix $\mathbf{\Psi} = (\mathbf{U}_{\text{Tx}})^T (\mathbf{A}_{\text{Tx}})^* \otimes (\mathbf{U}_{\text{Rx}})^H \mathbf{A}_{\text{Rx}}$

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- reshape vectorized estimate into matrix form: $\bar{\mathbf{H}}_a[n] = \text{vec}^{-1}(\bar{\mathbf{h}}_a[n])$

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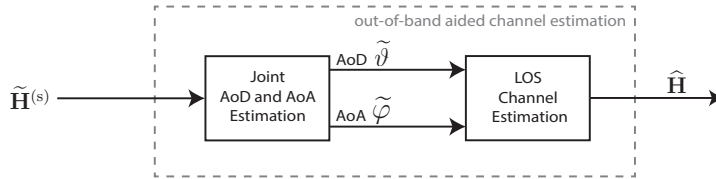
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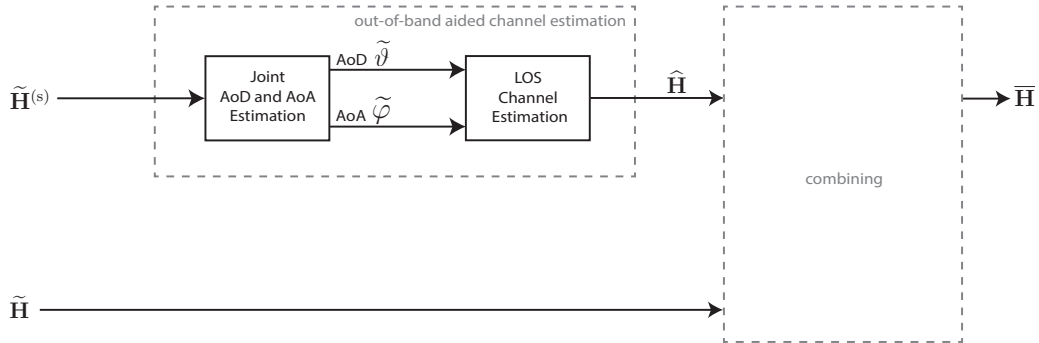
- reshape vectorized estimate into matrix form: $\bar{\mathbf{H}}_a[n] = \text{vec}^{-1}(\bar{\mathbf{h}}_a[n])$
- reconstruct desired quantized channel estimate: $\bar{\mathbf{H}}[n] = \mathbf{A}_{\text{Rx}} \bar{\mathbf{H}}_a[n] (\mathbf{A}_{\text{Tx}})^H$

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Channel Estimation Methods - Out-of-band Aided Maximal Ratio Combining (OOBA-MRC)



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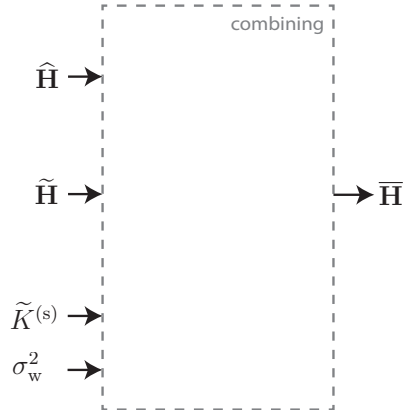


- gives an optimum performance for each combination of K -factor and noise variance [2]

$$\overline{\mathbf{H}}[n] = \hat{w} \hat{\mathbf{H}}[n] + (1 - \hat{w}) \tilde{\mathbf{H}}[n]$$

$$\hat{w} \approx \frac{M_{\text{Tx}} M_{\text{Rx}} \sigma_w^2}{\frac{M_{\text{Tx}} M_{\text{Rx}}}{1 + \tilde{K}^{(s)}} + (1 + M_{\text{Tx}} M_{\text{Rx}}) \sigma_w^2}$$

estimated sub-6 GHz K -factor $\tilde{K}^{(s)}$
mmWave noise variance σ_w^2



[2] F. Pasic, et al., "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information", IEEE Transactions on Communications (TCOM), 2025

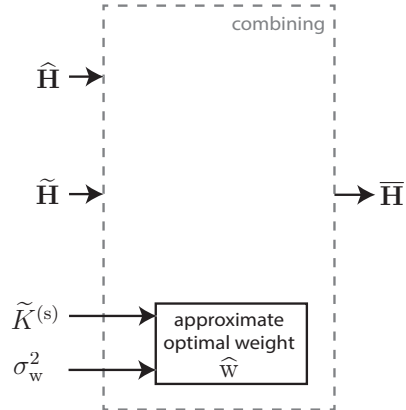
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[2] F. Pasic, et al., "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information", IEEE Transactions on Communications (TCOM), 2025

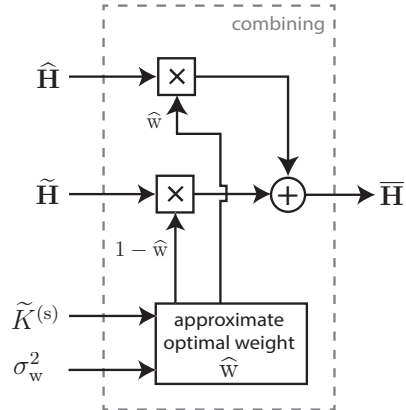
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Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
EIRP	30 dBm	30 dBm
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	120 kHz
Sampling Rate f_s	40.32 MHz	806.4 MHz
Cyclic Prefix t_{CP}	1.19 μs	0.59 μs
RF Impairments Power σ_{RF}^2	-25 dB	-25 dB
Mixer Insertion Loss IL_{mix}	7.5 dB	6 dB
Mixer Power Cons. P_{LO}	4 dBm	10 dBm
PA Power Added Eff. η_{PA}	25%	20%
LNA Power Cons. $P_{DC,LNA}$	7.05 mW	11 mW
DAC Figure of Merit FoM_{DAC}	65 fJ/conv	65 fJ/conv
ADC Figure of Merit FoM_{ADC}	67.6 fJ/conv	67.6 fJ/conv
ADC/DAC Resolution Bits n_b	10	{2, 4, 8}

